

BAYESIAN SAE WITH AN EMPHASIS ON LOW- AND MIDDLE-INCOME COUNTRIES: LECTURE 5: FURTHER TOPICS

Jon Wakefield

Departments of Statistics and Biostatistics
University of Washington

OUTLINE

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DISCUSSION

APPENDIX: STOCHASTIC PARTIAL DIFFERENTIAL
EQUATIONS

MOTIVATION

In these slides we discuss:

- ▶ Nested spatial models.
- ▶ Model-based geostatistics.
- ▶ Variance modeling in area-level models.
- ▶ Unmatched linking modeling in area-level models.
- ▶ The SAR model.
- ▶ Space-time modeling.
- ▶ Spatial machine learning.

NESTED SPATIAL MODELS

NESTED SPATIAL MODELS

The BYM2 spatial models we use smooth the raw data both **locally**, through the ICAR term, and **nationally**, through the IID term.

Oversmoothing is a concern when data are sparse, particularly for rare outcomes.

Nigeria has 774 Local Government Areas (LGAs) at Admin2, and SAE at this level using DHS or MICS is often treacherous.

Recall that surveys are powered to produce reliable estimates for some indicators at Admin1 and, hence, we would like to “recover” the weighted estimates, rather than distort away from those.

To try to reduce overshrinkage at the Admin1 level, we may use a **nested** model.

NESTED AREA-LEVEL MODEL

Let i denote Admin2 areas, $a[i]$ represent the Admin1 area within which Admin2 area i is contained, with $a = 1, \dots, A$, Admin1 areas.

A **nested Fay-Herriot area-level model** has the form:

$$\begin{array}{ccccccc} \hat{\theta}_i^w & | & \theta_i & \sim_{ind} & N(\theta_i, \hat{V}_i) \\ \theta_i & = & \underbrace{\alpha_{a[i]}}_{\text{Admin1 LEVEL}} & + & \underbrace{\mathbf{x}_i^T \boldsymbol{\beta}}_{\text{COVARIATE MODEL}} & + & \underbrace{u_i}_{\text{SPATIAL RESIDUAL}} \end{array} .$$

The terms α_a are Admin1 level parameters that are included to effectively recover the weighted estimates.

The u_i terms then spatially smooth **within** each Admin1 area – they are **nested random effects**.

This model can be viewed as being consistent with the sampling scheme in which the Admin1 areas correspond to **sampling strata**.

NESTED CLUSTER-LEVEL MODEL

- ▶ Y_{ic} deaths from n_{ic} births in **sampled cluster c** in **Admin2 area i** .

- ▶ **Overdispersed binomial** model:

$$Y_{ic} \mid r_{ic} \sim \text{BetaBinomial}(n_{ic}, r_{ic}, d), \quad (1)$$

where r_{ic} is the risk of neonatal death in cluster c of area i .

- ▶ **Nested model** for cluster risk p_{ic} :

$$\frac{r_{ic}}{1 - r_{ic}} = \exp\left(\underbrace{\alpha_{a[c]}}_{\text{Admin1 LEVEL}} + \underbrace{\mathbf{x}_{ic}^T \boldsymbol{\beta}}_{\text{COVARIATE MODEL}} + \underbrace{u_i}_{\text{SPATIAL RESIDUAL}} \right).$$

The terms α_a are Admin1 level parameters that are included to effectively recover the weighted estimates.

- ▶ The u_i terms then spatially smooth **within** each Admin1 area.
- ▶ This model can be viewed as being consistent with the sampling scheme in which the Admin1 areas correspond to **sampling strata**.

AN EXAMPLE WITH NESTED MODELS

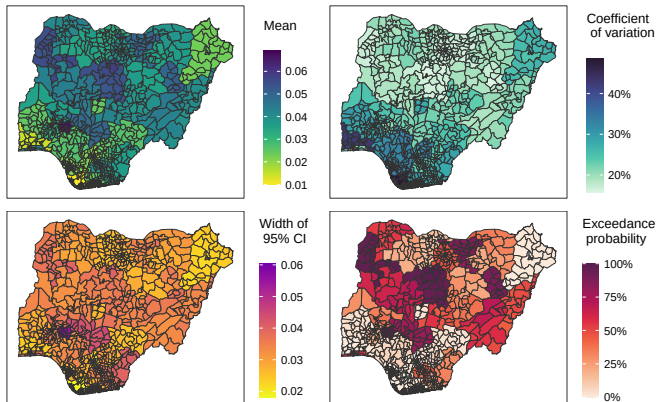


FIGURE 1: Fixed effects at Admin1, with nested BYM2 random effects at Admin2.

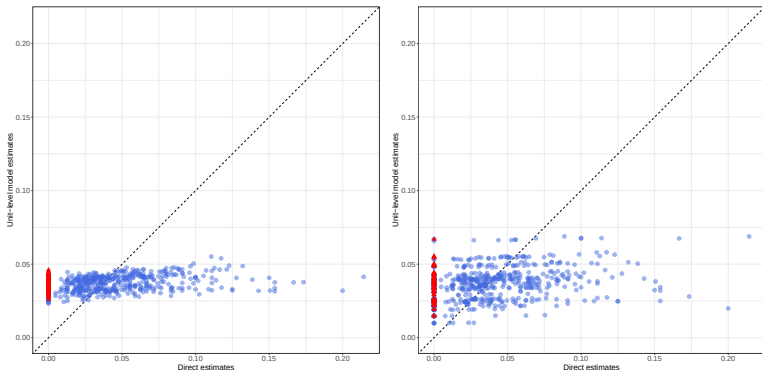


FIGURE 2: Estimates from non-nested (left) and nested (right) models, against weighted estimates. Notice the wider range in the right hand plot – the shrinkage has been **reduced**. But we don't pick up much variation within Admin1 areas, because the data are so sparse.

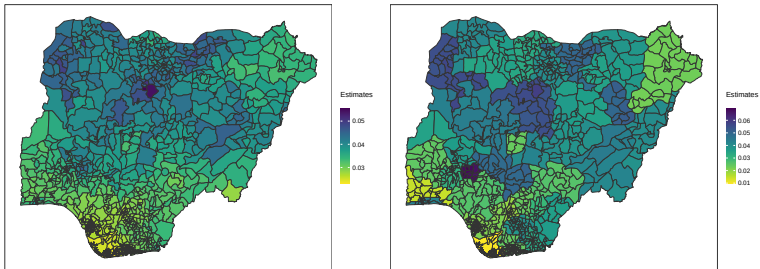


FIGURE 3: Mapped estimates from non-nested (left) and nested (right) models. Notice the larger scale in the right hand plot.

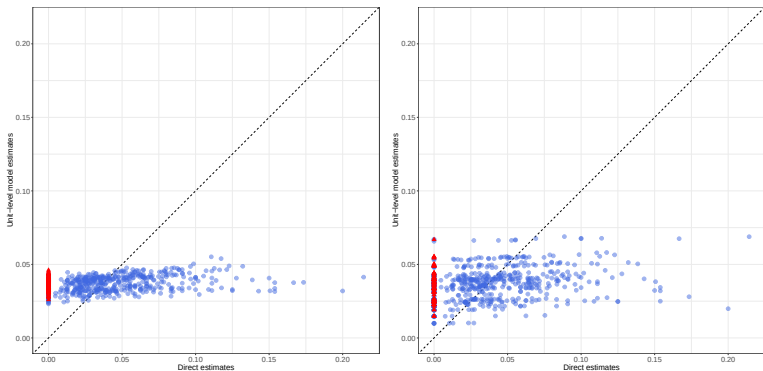


FIGURE 4: Estimates from non-nested (left) and nested (right) models, against weighted estimates. Notice the wider range in the right hand plot.

MODEL-BASED GEOSTATISTICS

MODEL-BASED GEOSTATISTICS (MBG)

- ▶ MBG models are (at first glance) unit-level models, with continuous spatial priors.
- ▶ The **sampling model** can be a binomial with a normal random effect (sometimes vaguely referred to as the nugget) on the logistic scale.
- ▶ Here we describe a **Betabinomial** model which is an example of an overdispersed binomial:

$$Y_{ic} \mid r_{ic}, d \sim \text{BetaBinomial}(n_{ic}, r_{ic}, d), \quad (2)$$

where $r_{ic} = r(\mathbf{s}_{ic})$ is the risk associated with location $\mathbf{s}_{ic}$.

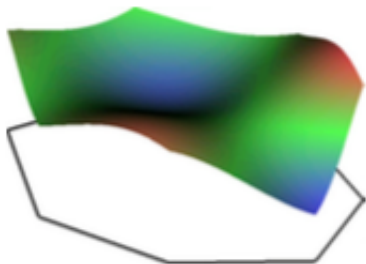
MODEL-BASED GEOSTATISTICS (MBG)

- ▶ The latent model is:

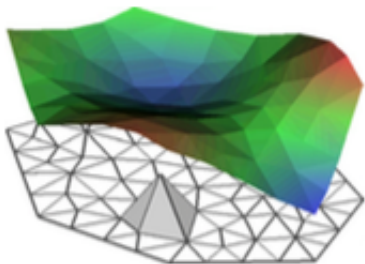
$$\frac{r(\mathbf{s}_{ic})}{1 - r(\mathbf{s}_{ic})} = \exp\left(\underbrace{\alpha}_{\text{OVERALL LEVEL}} + \underbrace{\mathbf{x}(\mathbf{s}_{ic})^T \beta}_{\text{COVARIATE MODEL}} + \underbrace{\omega(\mathbf{s}_{ic})}_{\text{SPATIAL RESIDUAL}} \right)$$

where $\omega(\mathbf{s}_{ic})$ is:

- ▶ Modeled as a realization from a **Gaussian Random Field (GRF)**.
- ▶ A **Matérn correlation function** is used, and implementation is via the Stochastic Partial Differential Equation (SPDE) approximation (Lindgren *et al.*, 2011) and implemented in R-INLA.
- ▶ We can replace α with a fixed effect $\alpha_{a[i]}$ where a indexes Admin1 units – this may aid in preventing **overshrinkage**.



(a) A continuous function.



(b) A piecewise linear approximation.

Fig. 2. Piecewise linear approximation of a function over a triangulated mesh.

FIGURE 5: GMRF representation of a Markovian GRF, via triangulation, from Simpson *et al.* (2012b).

AGGREGATION FOR MODEL-BASED GEOSTATISTICS

Aggregated risk, under the model, and assuming the sampled fraction of units in the area is negligible, is

$$r_i = \frac{1}{N_i} \sum_{k=1}^{N_i} r_{ik}$$

$\underbrace{\hspace{1.5cm}}_{\text{Modeling Approximation}} \approx \frac{1}{N_i} \sum_{k=1}^{N_i} \text{expit}(\alpha + \mathbf{x}^T(\mathbf{s}_{ik})\beta + \omega(\mathbf{s}_{ik})).$

Notice that, in contrast to the linear model case, we require the covariates on **each member of the population**.

In practice, we can aggregate over a grid:

$$r_i \approx \sum_{g=1}^{G_i} F_{ig} \times \text{expit}(\alpha + \mathbf{x}^T(\mathbf{s}_{ig})\beta + \omega(\mathbf{s}_{ig})), \quad (3)$$

with F_{ig} the fraction of the population units that are located in grid cell g , for example, from WorldPop population estimates (Tatem, 2017).

CONSTRUCTING A PREDICTIVE SURFACE

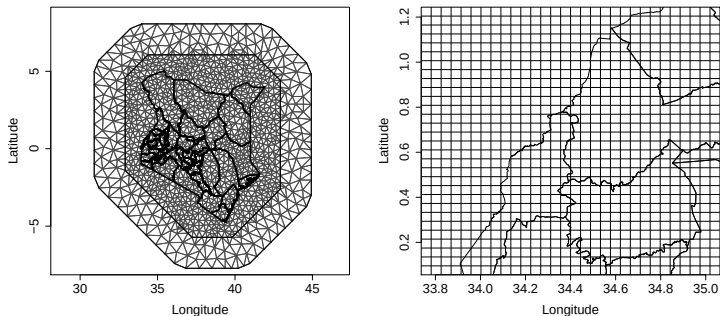


FIGURE 6: Inference via SPDE is carried out via the triangulation mesh (left), and then spatial predictions are calculated on a grid (right), as illustrated here for Kenya.

GEOSTATISTICAL MODELS

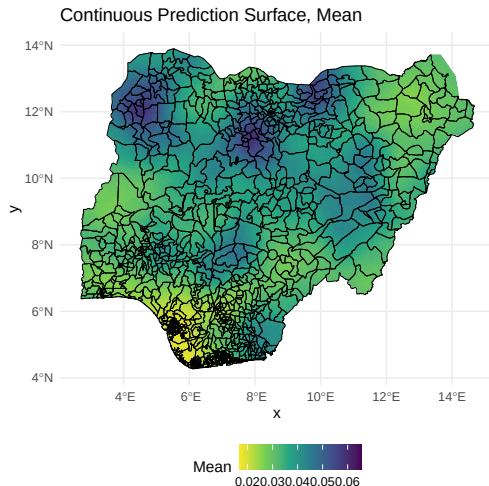


FIGURE 7: NMR continuous surface map for Nigeria, from 2018 DHS.

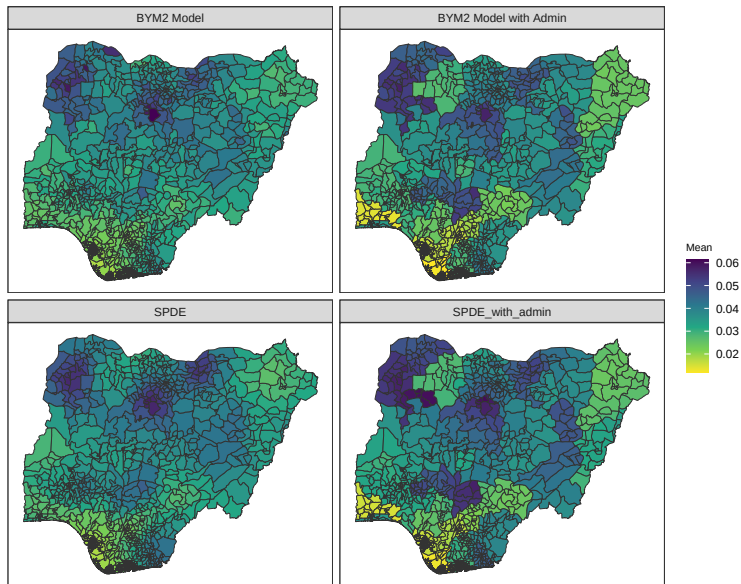


FIGURE 8: Comparison of aggregations from spatial models – BYM2 and GRF spatial models give quite similar area-level estimates.

VARIANCE FIX FOR FAY-HERRIOT

VARIANCE INSTABILITY

- ▶ The Fay-Herriot model is the most widely-used approach to SAE.
- ▶ But it requires as input the variance of the weighted estimate, $\hat{V}_i$, and this may be unstable or unavailable for certain configurations of data.
- ▶ In general, to alleviate sampling variance difficulties, generalized variance functions (Wolter, 2007, Chapter 7) may be used.
- ▶ Such approaches leverage the mean-variance relationship between θ_i^w and $\hat{V}_i^{1/2}$, and/or incorporate covariates (Otto and Bell, 1995; Mohadjer *et al.*, 2012; Franco and Bell, 2013; Liu *et al.*, 2014).
- ▶ We describe a simple approach that is implemented in our software.

VARIANCE MODELING

- ▶ For the surveys in LMICs the variance formula is:

$$\widehat{V}(\widehat{p}_i^w) = \frac{C_i}{C_i - 1} \frac{1}{(t_i^w)^2} \sum_{c \in S_i} \left[\sum_{k \in S_{ic}} w_{ick} (y_{ick} - \widehat{p}_{ic}^w) \right]^2 \quad (4)$$

- ▶ When the data are **sparse** in an area, then the estimator will have uncertainty that is high, or not possible to estimate (because the variance formula breaks down).
- ▶ Formula (4) breaks down if all outcomes in an area are all 0 or are all 1, or there is a single cluster in the area.
- ▶ We have a variance fix in `surveyPrev`, based on a weighted beta distribution.
- ▶ Alternatively, Gao and Wakefield (2023) jointly model weighted estimator and variance estimator.

IN THE BEGINNING...

Since the dawn of time, Bayesians have been fixing binomials with $y = 0/n$, e.g., Wilson (1927), combining binomial likelihood with $\text{Be}(2,2)$ prior.

$\text{Be}(2,2)$ = Phantom data with $z^{\text{PH}} = 2$ successes out of $m^{\text{PH}} = 4$ which gives the **augmented** estimator,

$$\hat{p}^{\text{AUG}} = \frac{z + z^{\text{PH}}}{m + m^{\text{PH}}}.$$

How to transfer over to the **survey setting**?

Consider the **pseudo-likelihood** for a generic area,

$$L^{\text{PL}}(p) = \prod_{k \in S} [p^{y_k} (1-p)^{1-y_k}]^{w_k}.$$

Now define a **pseudo prior**, based on phantom observations and weights, $\{y_k, w_k, k \in S^{\text{PH}}\}$,

$$g^{\text{PH}}(p) = \prod_{k \in S^{\text{PH}}} [p^{y_k} (1-p)^{1-y_k}]^{w_k} [p(1-p)]^{-1}.$$

This gives **augmented posterior**:

$$\pi^{\text{AUG}}(p) \propto \prod_{k \in S} [p^{y_k}(1-p)^{1-y_k}]^{w_k} \times \prod_{k \in S^{\text{PH}}} [p^{y_k}(1-p)^{1-y_k}]^{w_k}$$

Maximizing this augmented posterior gives estimator:

$$\hat{p}^{\text{AUG}} = \frac{\sum_{k \in S} w_k y_k + \sum_{k \in S^{\text{PH}}} w_k y_k}{\sum_{k \in S} w_k + \sum_{k \in S^{\text{PH}}} w_k}.$$

The augmented set of observations, $k \in S^{\text{AUG}}$, can be used in estimators and variance formulas.

This is implemented as default for pre-modeled estimates at

<https://sae4lmic.stat.uw.edu/>

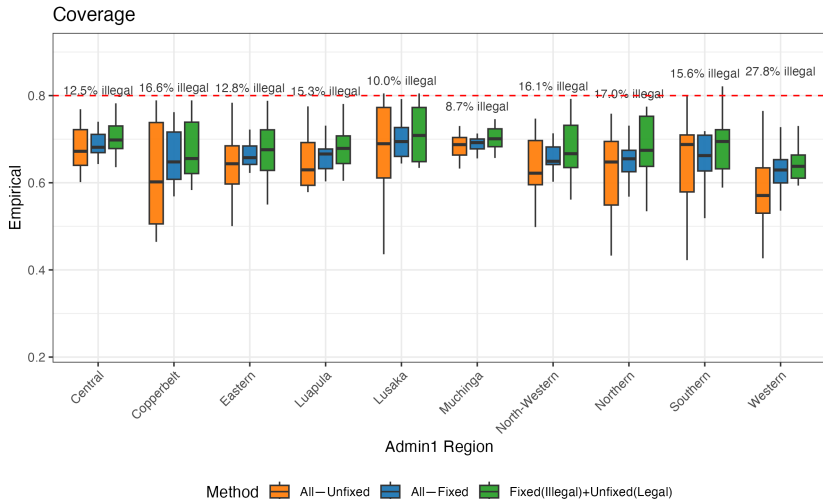


FIGURE 9: Empirical simulation coverage, mimicking DHS survey settings

UNMATCHED LINKING MODELS

UNMATCHED LINKING MODELS

- ▶ In the context of modeling a **prevalence**, we previously modeled on the **logit scale**.
- ▶ But $\text{logit}(\hat{p}_i^w)$ is not an **unbiased estimator** of the logit prevalence.
- ▶ To examine this, we consider the **unmatched linking model**:

$$\hat{p}_i^w \mid p_i \sim_{iid} N(p_i, \hat{V}_i^*), \quad (5)$$

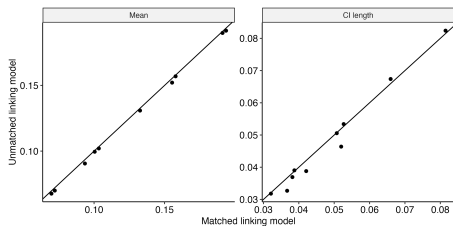
$$\text{logit}(p_i) = \alpha + u_i, \quad i = 1, \dots, m, \quad (6)$$

$$\mathbf{u} \mid \sigma_u^2, \phi \sim \text{BYM2}(\sigma^2, \phi), \quad (7)$$

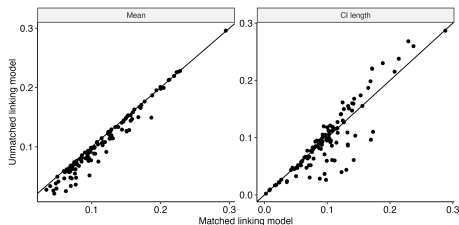
where $\hat{V}_i^*$ is the design-based variance estimate for $\hat{p}_i^w$.

UNMATCHED LINKING MODELS

- ▶ We fit this model to HIV prevalence data in adult women from the Zambia DHS, using the `surveyPrev` package.
- ▶ Figure 10 shows the resultant posterior mean and uncertainty estimates, plotted against estimates from the matched linking model.
- ▶ At the Admin1 level, we see strong agreement, which is reassuring, but there are systematic differences at the Admin2 level, as one might expect given the differences in the sampling models and the small sample sizes.



(a) Admin1 comparison.



(b) Admin2 comparison.

FIGURE 10: Comparison of posterior mean (left column) and 95% credible interval length (right column) under Fay-Herriot unmatched and matched linking models.

UNMATCHED LINKING MODELS

- ▶ Via linearization, one can show that there is a positive bias when the logit is modeled for $p_i < 0.5$, and we see this in the bottom left panel of Figure 10.
- ▶ In Figure 11, we plot the weighted national estimate of female adult HIV prevalence, along with aggregated national estimates from the matched and unmatched BYM2 models (with aggregation fractions in both cases taken as normalized WorldPop population counts).
- ▶ We see that relative to the weighted estimate, the unmatched estimate is low, whereas the matched model shows greater agreement.
- ▶ The problem with the unmatched model is that there are many areas with low prevalence estimates and in these cases the true sampling distribution is skewed, which the logit matched model picks up more effectively.

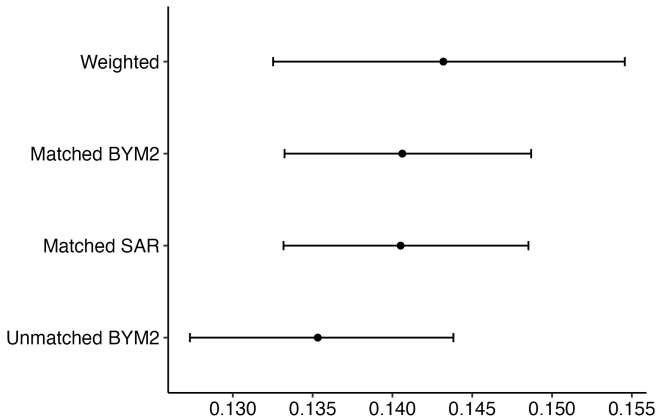
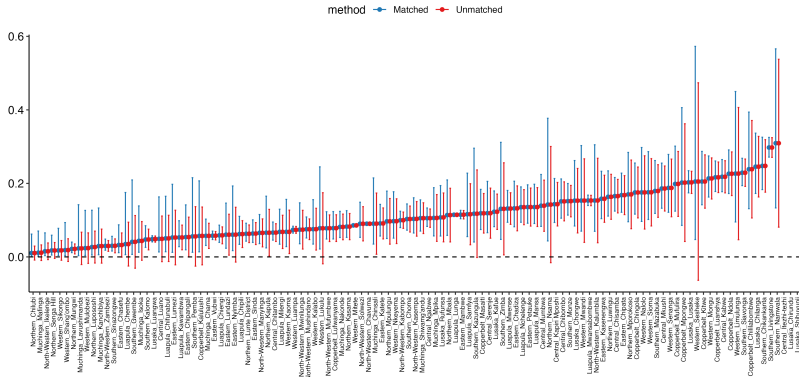


FIGURE 11: Comparison of weighted national estimate with estimates from various Fay-Herriot models, along with 95% uncertainty intervals.

UNMATCHED LINKING MODELS

- ▶ Figure 12 shows the point estimates and asymptotic normal 95% confidence intervals for the prevalence, for the original and logit scales.
- ▶ On the original scale (corresponding to the unmatched model) there are many areas with intervals that include negative values (around 20% of the areas).
- ▶ The assumed sampling distribution places excess mass on values close to zero, which pulls down the resultant estimates and leads to the negative bias in the aggregated national estimate.
- ▶ We conclude that, in this example, the matched model is preferred, but further research on its general use is merited.



THE SAR MODEL

SIMULTANEOUS AUTOREGRESSIVE (SAR) MODEL

We have so far described the BYM2 model for spatial smoothing, which models spatial dependence via an intrinsic conditional autoregressive (ICAR) form.

We now describe, as an alternative, a **simultaneous autoregressive (SAR)** Fay-Herriot model – this model is also available in the `surveyPrev` package.

It is also in the `sae` package (Marhuenda *et al.*, 2013), where inference is frequentist.

THE SAR MODEL

The SAR model (Banerjee *et al.*, 2015) is given by,

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \sum_{j=1}^n B_{ij}(y_j - \mathbf{x}_j^T \boldsymbol{\beta}) + \epsilon_i,$$

for $i = 1, \dots, m$, where

- ▶ the SAR weights B_{ij} have $B_{ii} = 0$,
- ▶ ϵ_i are independent, zero mean errors terms with variance $\hat{V}_i$,
- ▶ Let $\text{var}(\epsilon) = \hat{\mathbf{V}}$ be the $m \times m$ diagonal matrix with i -th element $\hat{V}_i$.

Conditional autoregressive (CAR) models (including the ICAR) proceed by conditioning on neighbors while the SAR model, as it says on the tin, considers simultaneous modeling.

Wall (2004) contains an interesting comparison of CAR and SAR models.

THE SAR MODEL

In vector form:

$$\mathbf{y} - \mathbf{x}^\top \boldsymbol{\beta} = \mathbf{B}(\mathbf{y} - \mathbf{x}^\top \boldsymbol{\beta}) + \boldsymbol{\epsilon},$$

or

$$(\mathbf{I}_m - \mathbf{B})(\mathbf{y} - \mathbf{x}^\top \boldsymbol{\beta}) = \boldsymbol{\epsilon},$$

or

$$\mathbf{y} = \mathbf{x}^\top \boldsymbol{\beta} + (\mathbf{I}_m - \mathbf{B})^{-1} \boldsymbol{\epsilon},$$

where the latter assumes that $\mathbf{B}$ has been chosen so that $\mathbf{I}_m - \mathbf{B}$ is invertible.

If $\mathbf{B} = \mathbf{0}$ we have an iid model.

So **spatial dependence** is induced through $(\mathbf{I}_m - \mathbf{B})^{-1}$.

Under normality of the errors

$$\mathbf{Y} \mid \beta, \mathbf{B} \sim \mathcal{N} \left(\mathbf{x}^\top \beta, [(\mathbf{I}_m - \mathbf{B})^\top \hat{\mathbf{V}} (\mathbf{I}_m - \mathbf{B})^{-1}]^\top \right).$$

For a well-defined model we require $(\mathbf{I}_m - \mathbf{B})$ to be non-singular, which puts conditions on $\mathbf{B}$.

A common choice is

$$\mathbf{B} = \rho_s \mathbf{W},$$

for a spatial proximity matrix $\mathbf{W}$ with elements 1 for neighbors and 0 otherwise.

THE SAR MODEL: SOME TECHNICAL DETAILS

Let $\lambda_{(1)}, \dots, \lambda_{(n)}$ be the ordered eigenvalues of $\mathbf{W}$.

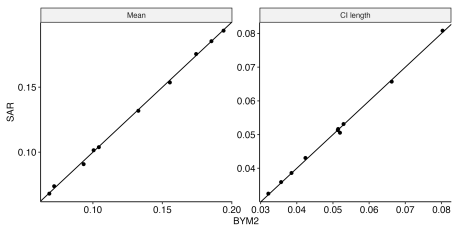
Then, $(\mathbf{I}_m - \mathbf{B})$ will be invertible if

$$\frac{1}{\lambda_{(1)}} < \rho_s < \frac{1}{\lambda_{(n)}}.$$

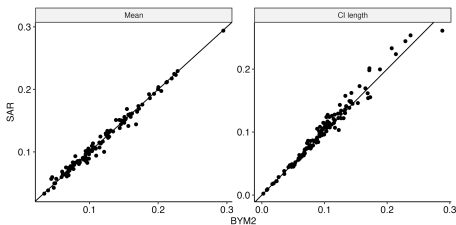
If the row sums of $\mathbf{W}$ are standardized to 1 then $\lambda_{(n)} = 1$ and $\lambda_{(1)} \leq -1$ so $\rho_s < 1$ but may be less than -1.

For more discussion, see Banerjee et al. (2015, Section 4.4) and Schabenberger and Gotway (2005, Section 6.2.2.1).

- ▶ We compare simultaneous autoregressive (SAR) and BYM2 Fay-Herriot models using the Zambia HIV DHS data.
- ▶ Specifically, we fit models at the Admin1 and Admin2 levels, using INLA (Gómez-Rubio *et al.*, 2020), obtaining almost identical point and interval estimates for the two models, as we see in Figure 13.
- ▶ Figure 11 illustrates that the aggregate national estimates under SAR and BYM2 models are also very similar.
- ▶ The SAR model and the unmatched model described by (5)–(7), are included in the `surveyPrev` package (Dong *et al.*, 2025).



(a) Admin1 comparison.



(b) Admin2 comparison.

FIGURE 13: Comparison of posterior mean (left column) and 95% credible interval length (right column) under Fay-Herriot models with SAR and BYM2 spatial random effects.

SPATIO-TEMPORAL MODELING

MAIN EFFECTS AND INTERACTIONS

- ▶ To motivate space-time models, when space is modeled discretely, we consider simple **two-way factor models**.
- ▶ Suppose we have a univariate **continuous** response Y .
- ▶ Now consider modeling as a function of two factors, A and B say, with $i = 1, \dots, m$ and $j = 1, \dots, T$ indexing the levels.
- ▶ A **main effects only model** takes the form

$$E[Y_{ij} \mid \alpha, \eta_i, \delta_j] = \alpha + \eta_i + \tau_j.$$

- ▶ **Interpretation:** η_i is the effect of being at level i for factor A, regardless of the level assumed by B, and τ_j is the effect of being at level j for factor B, regardless of the level assumed by A, i.e. there is no **interaction**.

MAIN EFFECTS AND INTERACTIONS

- ▶ An **interaction model** adds a set of interaction parameters

$$E[Y_{ij} \mid \alpha, \eta_i, \tau_j, \delta_{ij}] = \alpha + \eta_i + \tau_j + \delta_{ij}.$$

- ▶ **Interpretation:** δ_{ij} is the additional effect, beyond $\eta_i + \tau_j$ of being simultaneously at levels i and j of factors A and B.
- ▶ If the factor correspond to **nominal** levels (e.g., a factor for color with 2 levels: "red", "blue") then we would not expect similarity between adjacent levels.
- ▶ In a space-time context the "factors" **space** and **time** have an "ordering" and we might expect similarity in the response at neighboring levels.

MAIN EFFECTS MODEL

- ▶ First, consider the **space-time model**,

$$\begin{aligned} Y_{it} \mid \theta_{it} &\sim \text{N}(\theta_{it}, \widehat{V}_{it}) \\ \theta_{it} &= \text{expit}(\alpha + \eta_i + S_i + \omega_t + \tau_t) \end{aligned}$$

- ▶ Components:

- ▶ Y_{it} is the weighted estimate with design-based variance $\widehat{V}_{it}$.
- ▶ **Unstructured spatial term** $\eta_i \sim_{iid} \text{N}(0, \sigma_\eta^2), i = 1, \dots, m$.
- ▶ **Smooth spatial term** $[S_1, \dots, S_m]$, e.g., from an **ICAR model**.
- ▶ **Unstructured temporal term** $\omega_t \sim_{iid} \text{N}(0, \sigma_\omega^2), t = 1, \dots, T$.
- ▶ **Smooth temporal term** $[\tau_1, \dots, \tau_T]$ smooth in time, e.g. follows an **RW1 or RW2 model**.
- ▶ Notice there is **no interaction** between space and time – the spatial effects are constant across time and temporal trends are constant across space.

SPACE-TIME INTERACTION MODELS

- ▶ Knorr-Held (2000) considered the model:

$$\theta_{it} = \alpha + \eta_i + S_i + \omega_t + \tau_t + \delta_{it},$$

with η_i , S_i , ω_t , δ_t are as in the main effects only model.

- ▶ Four different models for the interaction δ_{it} :
 - ▶ **Type I:** Independent interaction.
 - ▶ **Type II:** Temporal trends differ between areas but don't have spatial structure.
 - ▶ **Type III:** Spatial patterns differ between time points but don't have temporal structure.
 - ▶ **Type IV:** Temporal trends differ between areas but more likely to be similar for adjacent areas.

We describe the Type IV model only, since it is the most appealing in a prevalence mapping context.

INSEPARABLE SPACE-TIME INTERACTION MODELS

- ▶ **Type IV:** Temporal trends differ between areas but more likely to be similar for neighboring areas.
- ▶ This will often be the most realistic model if interactions are present.
- ▶ In the case of a RW2 temporal model and an ICAR spatial model, the joint distribution can be written:

$$p(\delta \mid \sigma_\delta^2) \propto \exp \left(-\frac{1}{2\sigma_\delta^2} \sum_{t=3}^T \sum_{i \sim j} (\delta_{it} - \delta_{jt} - 2\delta_{i,t-1} + 2\delta_{j,t-1} + \delta_{i,t-2} - \delta_{j,t-2})^2 \right)$$

- ▶ The Knorr-Held (2000) models are implemented in the SUMMER package.

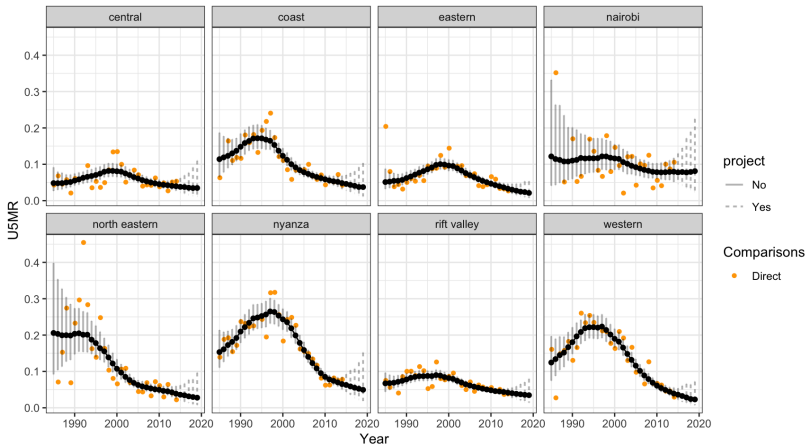


FIGURE 14: Weighted estimates and smoothed fits over time for 8 provinces of Kenya.

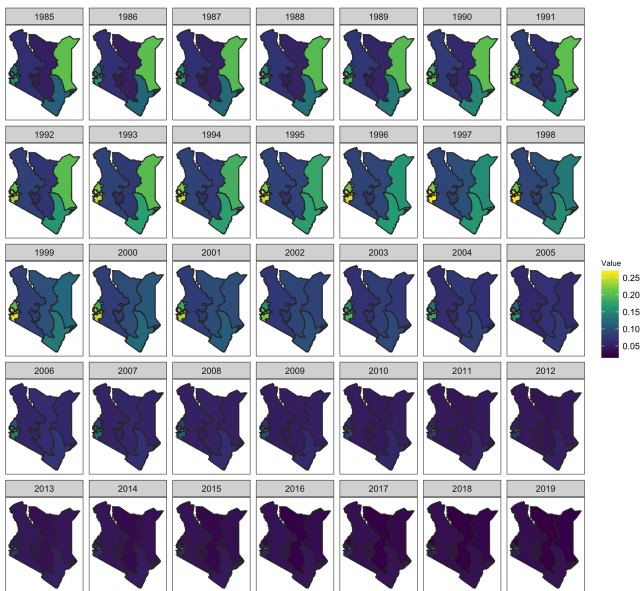


FIGURE 15: Maps of smoothed estimates over time for 8 provinces of Kenya.

Compare space-time interaction random effects

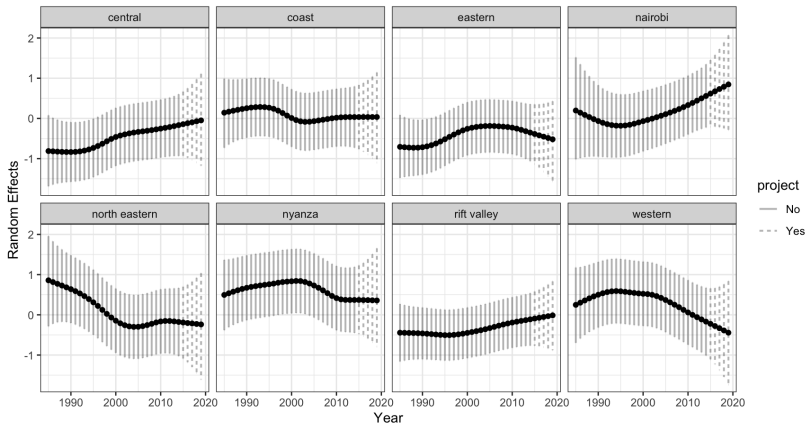


FIGURE 16: Space-time interactions δ_{it} for 8 provinces of Kenya.

SPACE-TIME FAY-HERRIOT MODEL (LI *et al.*, 2019)

- ▶ The **space-time Fay-Herriot** model has been used for 35 African countries to estimate U5MR in Admin1 regions, by year.
- ▶ Data enter at the 5-year level (to give stable variances, $\hat{V}_{it}$), but the RWs are defined on the 1-year scale.
- ▶ **Data:**
 - ▶ 121 DHS in 35 countries
 - ▶ 1.2 million children
 - ▶ 192 million child-months
- ▶ It took around 2.5 hours to obtain estimates for all countries – separate models for each country.
- ▶ This **area-level Fay Herriot model** is very reliable for examining Admin1 subnational variation, but the direct estimates are often unreliable for Admin2 estimation.

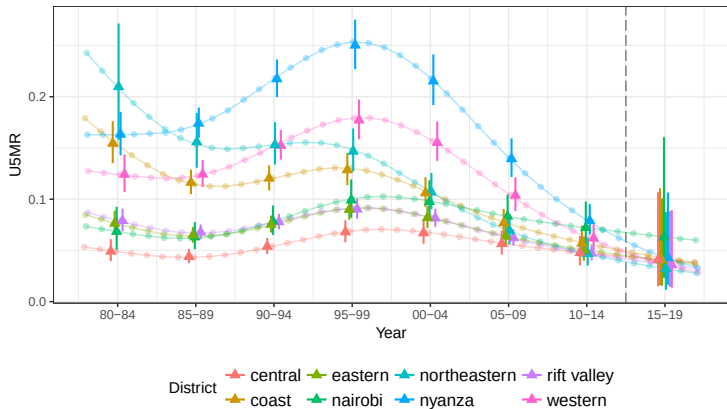


FIGURE 17: Posterior median estimates for Kenya districts.

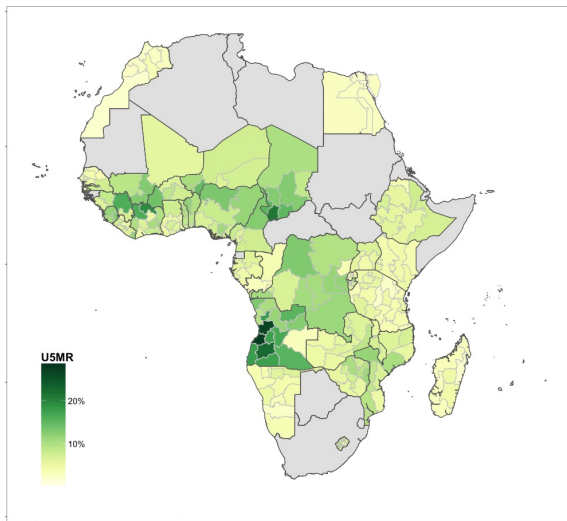


FIGURE 18: Predictions of U5MR for 2015, in 35 countries of Africa.

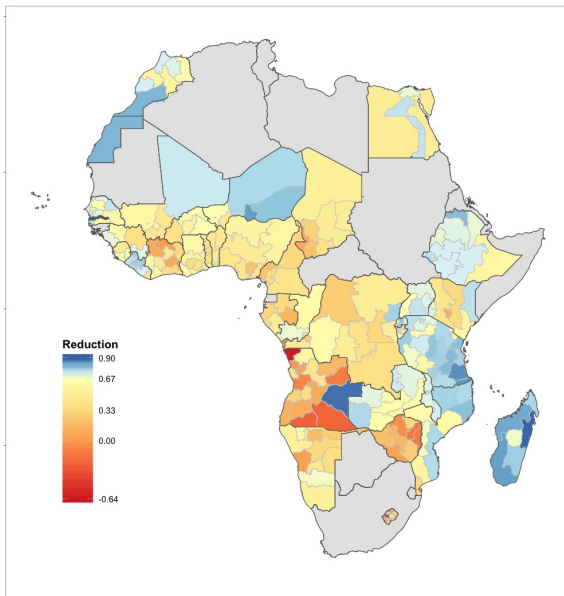


FIGURE 19: Percent reduction from 1990 to 2015, in 35 countries of Africa.

SPATIAL MACHINE LEARNING

THE PLEASURE AND THE PAIN

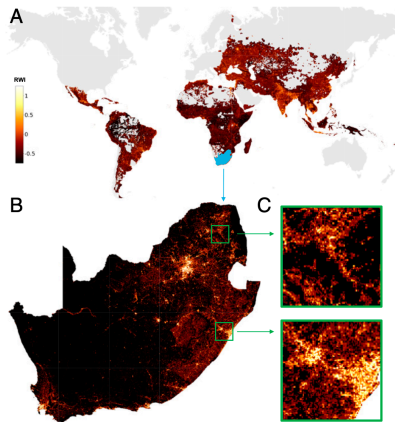


Fig. 1. Microestimates of wealth for all low- and middle-income countries. (A) Estimates of the relative wealth of each populated 2.4-km gridded region of all 135 LMICs. Interactive version is available at <http://www.povertymaps.net>. (B and C) Enlargements show (B) the countries of South Africa and Lesotho and (C) the regions around Durban and Polokwane.

ML approaches are very appealing:

- ▶ Flexible mean function to allow interactions and nonlinearities.
- ▶ Can be used with a large number of covariates (census, facilities, climatic, satellite imagery, social media, cell phone,...).
- ▶ Automatic model selection (hmmm).

But:

- ▶ How to tune algorithm?
- ▶ How to account for weights, spatial dependence.
- ▶ How to obtain measures of uncertainty?
- ▶ Lots of suggestions, but where is the theory?

FIGURE 20: From Chi et al. (2022).

UNCERTAINTY ESTIMATION IN SPATIAL ML

- ▶ Many different resampling algorithms have been used: the bootstrap, jackknife, conformal prediction, prediction powered inference (**PPI**),...
- ▶ Model-assisted estimator:

$$\hat{T}_i = \sum_{k \in U_i} \hat{\mu}(\mathbf{x}_{ik}) + \sum_{k \in S_i} \frac{y_{ik} - \hat{\mu}(\mathbf{x}_{ik})}{\pi_{ik}}$$

- ▶ Leverages population-level information.
- ▶ Links with many literatures including AIPW and casual inference, popular in the survey literature since the 1970s (Cassel *et al.*, 1976; Särndal *et al.*, 1992).
- ▶ Rediscovered to form the basis for **PPI** (Angeloulous et al, 2023).
- ▶ Very large sample size are needed, and the same is true for conformal prediction.
- ▶ Gao and Wakefield (2024) use model-assisted models in SAE.
- ▶ Dharamshi *et al.* (2025) use U/V statistic representations of model-assisted to obtain exact variance estimates that account for uncertainty in $\hat{\mu}$.

DISCUSSION

DISCUSSION

- ▶ There are many discrete spatial smoothing models, and it may not make much difference which is used, in terms of predictions and uncertainty – so pick the one you're comfortable with.
- ▶ Continuous spatial modeling, as used in MBG, is appealing, but aggregation is much more difficult, and computation is greater.
- ▶ Continuous spatial models may be combined with temporal models but it gets quite complex and computation is complex, but models can be fitted in INLA, see Wakefield *et al.* (2019).
- ▶ Beware machine-learning approaches that are magical to behold, but most approaches ignore the survey aspects, and do not give valid uncertainty estimates (the vanilla bootstrap is not valid for many ML techniques, especially in a spatial setting).
- ▶ Smoothing over space and time can be done within INLA, but computational expense increases steeply if a third variable (such as age) is added.

- ▶ Extensive materials can be found at

<https://sae4health.stat.uw.edu/>

- ▶ For space only, stat with surveyPrev, for space-time SUMMER.
- ▶ If you can't find something you need, write to me at jonno@uw.edu

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APPENDIX: STOCHASTIC PARTIAL DIFFERENTIAL EQUATIONS

The rationale behind the SPDE approach is:

1. GPs with Matérn covariance functions are the solutions to a particular **stochastic partial differential equation**.
2. These SPDEs can be numerically solved using an approximation, based on the **finite element method**, that produces a **sparse precision matrix**. Because of the sparseness, this approach is implemented in INLA.

In the following, we draw on Lindgren *et al.* (2011); Simpson *et al.* (2012a,b); Krainski *et al.* (2018); see also Miller *et al.* (2020) for a reasonably readable account – in particular, the .

There are two main parts to the SPDE approach to implementing GP models: represent the GP using an SPDE and solving using the finite element method.

The first part is based on a result of Whittle (1954) in which he showed that Matérn GPs can be represented as the solution to the SPDE:

$$(\kappa^2 - \Delta)^{\alpha/2} \tau f(\mathbf{x}) = W(\mathbf{x}), \quad (8)$$

where

- ▶ $\Delta = \sum_{j=1}^d \frac{\partial^2}{\partial x_j^2}$ is the Laplacian,
- ▶ $\alpha = \nu + d/2 > 0$ where ν is the smoothing parameter and d is the dimension of $\mathbf{x}$.
- ▶ $W(\mathbf{x})$ is white noise with unit variance.
- ▶ τ^2 is related to the variance.

The solution to (8) corresponds to a stationary GP for f with a **Matérn covariance function**:

$$\text{cov}[f(\mathbf{x}_i), f(\mathbf{x}_j)] = \frac{\sigma^2}{\Gamma(v + 1/2)\sqrt{4\pi\kappa^{2v}2^{v-1}}}(\kappa \|\mathbf{x}_i - \mathbf{x}_j\|)^v K_v(\kappa \|\mathbf{x}_i - \mathbf{x}_j\|)$$

where $K_v(\cdot)$ is a modified Bessel function of the second kind, $\sigma^2 > 0$ is a variance parameter, $\kappa > 0$ is a scale parameter and $v > 0$ is a smoothness parameter.

The variance is related to τ^2 via,

$$\sigma^2 = \frac{\Gamma(v)}{\Gamma(\alpha)(4\pi)^{d/2}\kappa^{2v}\tau^2}$$

MARKOVIAN IN CONTINUOUS SPACE

If α is an integer then the GRF has the **continuous Markov property** (Whittle, 1954), which is the key for implementation.

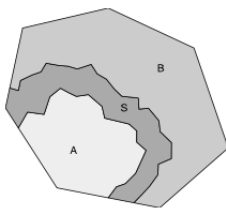


FIGURE 21: From Simpson *et al.* (2012b) who present this figure as an illustration of the spatial Markov property. If $\{x(\mathbf{x}) : \mathbf{x} \in A\}$ is independent of $\{x(\mathbf{x}) : \mathbf{x} \in B\}$ given $\{x(\mathbf{x}) : \mathbf{x} \in S\}$, then the field $x(\mathbf{x})$ has the **spatial Markov property**.

A BIT OF INTUITION ON THE SPDE

For $d = 1$ imagine we discretize time with equally intervals h :

$$-\frac{d^2 f}{dx^2} \approx \frac{1}{h^2} [2f(x) - f(x-h) - f(x+h)],$$

which is a SAR model

$$2f(x_i) - f(x_{i-1}) - f(x_{i+1}) \sim N(0, h^2 \sigma^2).$$

So if we let the spacing $h \rightarrow 0$ we get

$$-\Delta f(x) = -\frac{d^2 f}{dx^2} \sim W(x).$$

Similarly, we can discretize space on a grid with a mesh that has side length h , then

$$\begin{aligned} -\frac{d^2 f}{dx^2} &\approx \frac{1}{h^2} [2f(x_1, x_2) - f(x_1 - h, x_2 - h) - f(x_1 - h, x_2 + h) \\ &\quad - f(x_1 + h, x_2 - h) - f(x_1 + h, x_2 + h)] \end{aligned}$$

So SPDEs are the limit of a certain type of SAR model.

MATÉRN MEMBERS THAT SATISFY THE CONTINUOUS MARKOV PROPERTY

In 1D (and other **odd** dimensions) $\nu = 1/2, 3/2, \dots$, satisfy the continuous Markov property:

- ▶ Thin plate splines correspond to $\nu = 1$.
- ▶ The exponential covariance model corresponds to $\nu = 1/2$.

For 2D (and other **even** dimensions), the continuous Markov property is satisfied for $\nu = 1, 2, 3, \dots$:

- ▶ Thin plate splines correspond to $\nu = 1$.
- ▶ The exponential covariance model corresponds is not Markovian in 2D.

Discretizations of space lead to straightforward definitions of being Markovian. Not so easy in continuous space, but intuitively as below, but how can we use this for computation?

So we know that certain Matérn fields have the continuous Markov property, but unfortunately we cannot immediately write down a sparse precision matrix and use the Rue/Held tricks for an efficient implementation.

In the lingo, not every approximation to the precision operator will be sparse; in particular the approximation you'd get by evaluating the covariance matrix and inverting it won't usually be sparse.

Similarly the approximation you get using the Karhunen-Loève expansion¹ won't be sparse (Simpson *et al.*, 2012b).

Another way of seeing this: if you have a GMRF on a grid and you make a new multivariate Gaussian by locally averaging the GRMF, the resulting Gaussian doesn't have a sparse precision matrix

¹a representation of a stochastic process as an infinite linear combination of orthogonal functions, analogous to a Fourier series representation of a function on a bounded interval

FINITE ELEMENT APPROACH

We need to construct a solution to the SPDE, in the case where the observations are not observed on a regular grid, but are irregularly spaced.

We want to find an approximation,

$$f(\mathbf{x}) \approx \sum_{k=1}^m w_k \phi_k(\mathbf{x})$$

that best approximates the SPDE.

In the approximation the w_k are random (multivariate Gaussian) and $\phi_k(s)$ are a set of basis functions, $k = 1, \dots, m$.

Lindgren *et al.* (2011) describe how to carefully choose the basis functions to preserve the sparse structure of the resulting precision matrix – this is done using the [finite element method](#), a common technique for solving sets of partial differential equations.

STOCHASTIC PARTIAL DIFFERENTIAL EQUATIONS (SPDEs)

For simplicity take $\alpha = 2$ and $d = 2$; then, as a reminder, we are trying to find the $f(\mathbf{x})$ that is the solution to the SPDE

$$(\kappa^2 - \Delta)\tau f(\mathbf{x}) = W(\mathbf{x}), \quad (9)$$

Any solution to (9) also satisfies, for any suitable $\phi(\mathbf{x})$,

$$\int_{\mathbb{R}^2} (\kappa^2 - \Delta)f(\mathbf{x})\tau\phi(\mathbf{x}) d\mathbf{x} = \int_{\mathbb{R}^2} \phi(\mathbf{x}) dW(\mathbf{x}). \quad (10)$$

In the FEM method the smooth test function $\phi(\mathbf{x})$ is replaced by a set of m well-chosen bases, the **finite elements** – this is the **Delauney triangulation** which gives functions $\phi_k(\mathbf{x})$ that are piecewise linear functions that are equal to 1 on vertex k and 0 on all other vertices – this is the **triangular mesh**.

The limited support is key to producing an approximation that has a sparse precision matrix.

STOCHASTIC PARTIAL DIFFERENTIAL EQUATIONS (SPDEs)

These finite element test functions lead to the linear equation

$$(\kappa^2 \tilde{\mathbf{C}} + \mathbf{G}) \tilde{\mathbf{f}} \stackrel{d}{=} \mathbf{N}(\mathbf{0}, \tilde{\mathbf{C}})$$

where the matrices are given by

- ▶ $[\tilde{\mathbf{C}}]_{i,j} = \int \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) d\mathbf{x}$
- ▶ $[\tilde{\mathbf{G}}]_{i,j} = \int \nabla \phi_i(\mathbf{x}) \cdot \nabla \phi_j(\mathbf{x}) d\mathbf{x}.$

These integrals can be explicitly calculated (Lindgren *et al.*, 2011) and are only non-zero if vertex i is a neighbor of vertex j – hence, these matrices are **sparse** (and symmetric positive definite), so very convenient for manipulation.

This leads to

$$\tilde{\mathbf{f}} \stackrel{d}{=} \mathbf{N}(\mathbf{0}, \tilde{\mathbf{Q}}^{-1}),$$

where

$$\tilde{\mathbf{Q}} = (\kappa^2 \tilde{\mathbf{C}} + \mathbf{G})^\top \tilde{\mathbf{C}}^{-1} (\kappa^2 \tilde{\mathbf{C}} + \mathbf{G}).$$

STOCHASTIC PARTIAL DIFFERENTIAL EQUATIONS (SPDEs)

Unfortunately $\tilde{\mathbf{C}}$ is not a diagonal matrix and so $\tilde{\mathbf{Q}}$ is not sparse.

Lindgren *et al.* (2011) replace $\tilde{\mathbf{C}}$ by the diagonal matrix $\mathbf{C}$ that has on its diagonal the row sums of $\tilde{\mathbf{C}}$.

This gives the approximating GMRF as

$$\mathbf{f} \stackrel{d}{=} \mathbf{N}(\mathbf{0}, \mathbf{Q}^{-1}),$$

where

$$\mathbf{Q} = (\kappa^2 \mathbf{C} + \mathbf{G})^\top \mathbf{C}^{-1} (\kappa^2 \mathbf{C} + \mathbf{G})$$

This gives a sparse precision matrix, so that computation can be carried out efficiently.

This method is available within the INLA package – see Section 2.2.2 of Krainski *et al.* (2018) where they walk through a numerical example showing the constituent matrices, and the details of the INLA implementation, including a description of the `inla.mesh.fem` function that constructs the finite elements.